

Homework for week 3, **submission deadline 23:59 on Sunday night.**

1. Use nature microbiology paper from last week and use a GSE identified from Week 2 homework, answer the question list in the table below. If you use AI-agent or R to facilitate this process, please paste code or script at the end.

Step	Student action	Evidence to record
1	State the Week 2 data-driven question.	One sentence with population/system, comparison, outcome, and data type.
2	Confirm study identity.	GSE, title, organism, publication, and platform.
3	Map accessions.	At least one GSM and any linked BioProject/SRP/SRR.
4	Identify available data layers.	Series matrix, supplementary counts, raw reads, or single-cell files.
5	Reconstruct the design.	Groups, n, subject/paired structure, major covariates, batch.

Step 1: State the Week 2 Data-Driven Question

Item	Evidence
Question	In patients with colorectal cancer (CRC), inflammatory bowel disease (IBD), and irritable bowel syndrome (IBS), are specific mucosal gut microbial taxa significantly associated with host colonic epithelial gene expression, and do these associations exhibit shared patterns across diseases as well as disease-specific differences?

Step 2: Confirm Study Identity

Item	Evidence
GSE (host RNA-seq)	GSE111889 (IBD cohort host RNA-seq)

Item	Evidence
GSE (16S rRNA)	Not in GEO; available via SRA BioProject PRJNA398089 (IBD cohort 16S rRNA)
Title	“Identification of shared and disease-specific host gene–microbiome associations across human diseases using multi-omic integration”
Organism	Homo sapiens (colonic mucosal biopsies)
Publication	Priya et al., Nature Microbiology, 2022; 7(6): 780–795. DOI: 10.1038/s41564-022-01121-z
Platform	Host RNA-seq: Illumina HiSeq 2500 (paired-end, 101 bp) ; 16S: Illumina MiSeq (V4 region, 2×250 bp)

Note: The paper used three disease cohorts (CRC, IBD, IBS). Each cohort has distinct accessions. The IBD cohort is uniquely suited for GSE-based reuse because its host transcriptomic data is deposited in GEO (GSE111889), while its microbiome data is in SRA (PRJNA398089).

Step 3: Map Accessions

Accession Type	Identifier	Notes
GSE (host RNA-seq)	GSE111889	IBD cohort host transcriptomic data
GSM (example)	GSM3043382	A representative sample accession within GSE111889
BioProject (16S rRNA)	PRJNA398089	IBD cohort mucosal 16S rRNA sequencing data
SRP/SRR (16S rRNA)	Within PRJNA398089	Raw sequence reads accessible via SRA
Additional (IBS host RNA-seq)	GSE146853	IBS cohort host transcriptomic data (if needed for comparison)

Step 4: Identify Available Data Layers

Data Layer	Availability	Accession/Location
Series matrix	Available	GSE111889 (GEO)
Supplementary counts	Available	GSE111889 supplementary files (gene count matrices)
Raw reads (host RNA-seq)	Available	SRA within GSE111889
Raw reads (16S rRNA)	Available	SRA BioProject PRJNA398089
Processed data tables	Available	Supplementary Tables S12–S17 of the paper (host transcriptomics and microbiome data for each cohort)
Single-cell files	Not available	This study used bulk RNA-seq, not single-cell

Step 5: Reconstruct the Design

Design Element	Evidence
Groups	IBD cohort: 56 patients with IBD (34 Crohn's disease + 22 ulcerative colitis) and 22 non-IBD controls; total n = 78 individuals
n	78 paired samples (host RNA-seq + 16S rRNA) from the IBD cohort; across all three cohorts: 208 paired samples (CRC: 88; IBD: 78; IBS: 42)
Subject/paired structure	Paired data: each individual contributed one microbiome sample and one host gene expression sample from colonic mucosal biopsies
Major covariates	Gender (male/female); disease subtype for IBD (CD vs. UC)
Batch	Cross-cohort batch effects were minimized via meta-analysis approach; analyses were performed separately within each cohort, and results were

Design Element

Evidence

combined at the final step

Code / Script Used to Facilitate This Process

Below is the R script used to query GEO for study identity and data layers (for reproducibility):

```
1 # Load required packages
2 library(GEOquery)
3 library(rentrez)
4
5 # Step 1: Fetch GSE111889 metadata from GEO
6 gse_ibd <- getGEO("GSE111889", GSEMatrix = TRUE)
7 show(gse_ibd) # View platform, sample count, etc.
8
9 # Step 2: Extract sample-level metadata (GSM accessions)
10 pdata <- pData(phenoData(gse_ibd[[1]]))
11 gsm_accessions <- rownames(pdata)
12 head(gsm_accessions) # e.g., GSM3043382
13
14 # Step 3: Query SRA for linked BioProject
15 # Note: GSE111889 host RNA-seq; 16S rRNA for IBD cohort is in PRJNA398089
16 sra_search <- entrez_search(db = "sra", term = "PRJNA398089")
17 sra_summary <- entrez_summary(db = "sra", id = sra_search$ids)
18 print(sra_summary)
19
20 # Step 4: List supplementary files
21 getGEOSuppFiles("GSE111889")
22 list.files("GSE111889")
```

Script source: Adapted from GEOquery and rentrez documentation for the IBD cohort accessions identified in the Nature Microbiology paper.

2. GSE87487 is a human bulk RNA-seq study of 20 liver allograft biopsies, with pre- and post-reperfusion sampling and a 3.7 MB supplementary count matrix. The Series links to SRA project SRP090633. If you use AI-agent or R to facilitate this process, please paste code or script at the end.

Question	Student decision
Are the values counts?	Verify integer values after download.
Are samples independent?	Preserve subject pairing in metadata and design.
Can all 20 samples be compared directly?	Define a specific contrast before analysis.
Is full raw-read reprocessing possible?	Possible, but beyond a one-hour homework.

Question

Student decision for GSE87487

Are the values counts?

Yes. The supplementary file `GSE87487_counts.20samples.txt.gz` is a featureCounts-derived **count matrix** (3.7 MB), and the study methods confirm reads were aligned to hg38 with STAR and counted with featureCounts. Verify integer values after download.

Are samples independent?

No. This is a **paired design** with pre- and post-reperfusion biopsies from the **same donor liver**. Preserve the subject/paired structure in metadata and use paired statistical models (e.g., paired t-test or mixed-effects model) rather than treating samples as independent.

Can all 20 samples be compared directly?

Not as a single unstratified comparison. The 20 samples comprise **pre- and post-reperfusion pairs** across IRI+ and IRI- patients. You must define a specific contrast (e.g., the **Time × IRI interaction**, $\Delta\Delta = (\text{Post-Pre})\text{IRI}+ - (\text{Post-Pre})\text{IRI}-$) before analysis, as performed in subsequent reuses of this dataset.

Is full raw-read reprocessing possible?

Technically yes (raw data are in **SRA project SRP090633**), but likely **beyond a one-hour homework**. The provided count matrix is sufficient for most downstream analyses; raw-read reprocessing would only be needed for specialized tasks like alignment QC or isoform-level quantification

```
1 # Load required packages
2 library(GEOquery)
3
```

```
4 # Fetch GSE87487 metadata
5 gse <- getGEO("GSE87487", GSEMatrix = TRUE)
6 show(gse)
7
8 # Extract sample-level metadata
9 pdata <- pData(phenoData(gse[[1]]))
10 print(pdata[, c("title", "characteristics_ch1", "source_name_ch1")])
11 print(table(pdata$title))
12
13 # Check supplementary files
14 getGEOSuppFiles("GSE87487")
15 list.files("GSE87487")
```